

About ideal structure of Weyl algebra and introduction to differential operators.

Xinyuan. Zhang

Let $K[x_1, \dots, x_n] := K[X]$. Denote $\hat{x}_i f = x_i \cdot f$ for $f \in K[x_1, \dots, x_n]$. The Weyl algebra A_n is defined as

$$A_n = \langle \hat{x}_1, \dots, \hat{x}_n, \partial_1, \dots, \partial_n \rangle$$

The relation is $[\partial_i, x_j] = \delta_{ij}$.

This algebra has a K -basis

$$x^\alpha \partial^\beta =$$

We begin with the ideal of A_n .

For any $D \in A_n$. $\deg(D) = |\alpha| + |\beta|$.

Theorem 0.1. $\forall D, D' \in A_n$, we have:

1. $\deg(D + D') \leq \max\{\deg(D), \deg(D')\}$.
2. $\deg(D \cdot D') = \deg(D) + \deg(D')$.
3. $\deg([DD']) \leq \deg(D) + \deg(D') - 2$.

Proof. 1

□

Theorem 0.2. A_n is simple (as associative algebra).

Proof. 2

□

Corollary 0.3. All non-zero elements in $\text{End}(A_n)$ are injective.

For characteristic $p > 0$.

1 Rings of differential operators

Let R be a commutative k -algebra. For any a in K Define map $v \mapsto av$. order 0-n.

Lemma 1.1. Let $D' = \text{Der}(R) + R$

Proof. 3

□

Definition 1.2. $D(R) = \bigcup$

Proposition 1.3. If $P \in D^n$ and $Q \in D^m$, then $PQ \in D^{m+n}$.

Proof. 4

□

Proposition 1.4. $\text{Der}(K[X])$

Proof. 5

□

We want to prove $A_n = D(K[X])$.

Lemma 1.5. $P \in D(K[X]), \forall i, [P, x_i] = 0$, then $P \in K[X]$.

Proof. 6

□

Definition 1.6. C_r

Lemma 1.7. For $P_1, \dots, P_n \in C_{r-1}$, $\forall [P_i, x_j] = [P_j, x_i]$

Proof. 7

□

Theorem 1.8. $D(K[X]) = A_n$

Proof. 8

□

2 A_n -modules

Definition 2.1. torsion elements

Definition 2.2. torsion module

Lemma 2.3. If M is an irreducible R -module.

1. $M \cong R/\text{Ann}(u)$.
2. If R is not a division ring, then M is a torsion module.

Proposition 2.4. $K[X]$ is an irreducible torsion A_n -module. Further more, $K[X] \cong A_n/\sum_{i=1}^n A_n \partial_i$.

Proof.

□

We also have $A_n/\sum A_n x_i \cong K[\partial_1, \dots, \partial_n]$.

Proposition 2.5. *Twisting R -mod M .*

Proposition 2.6. (1) M irr $\Leftrightarrow M$ twis irr (2) $\text{tor} \Leftrightarrow \text{twis tor}$ (3) J is a left ideal of R so is $\sigma(J)$.

Let $F : A_n \rightarrow A_n$, $x_i \mapsto \partial_i$, $\partial_i \mapsto -x_i$. We call M_F the F -transform of M .

Proposition 2.7. $(K[X])_F = K[\partial_1, \dots, \partial_n] = K[\partial]$